

# Exact Hazards and Residual Life for Twenty Frozen First-Passage Laws

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## Abstract

The frozen C88 receipt gives exact first-passage distributions for twenty subgroup targets under a uniform ordering of sixteen named labels. We derive the complete discrete hazard sequence for every target and, after every surviving prefix, the full conditional residual-life distribution, mean, and variance. All quantities are reduced rational numbers on a finite permutation space. A second implementation reconstructs the passage counts from the complete C88 hit bitsets before checking every hazard and residual cell. SymPy arithmetic, clean replay, and thirteen hostile mutations complete the audit. Empty conditioning events remain explicitly undefined. The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

## 1 Frozen model and conventions

Let  $L$  be the sixteen frozen labels, let  $\pi$  be uniformly distributed on the  $16!$  orderings of  $L$ , and let  $A_k(\pi)$  be its first  $k$  labels. For one of the twenty actual subgroup targets  $H$ , C88 defines

$$T_H = \min\{k : H \leq \Phi(A_k)\}, \quad 0 \leq T_H \leq 16.$$

Write

$$N_H(k) = \#\{\pi : T_H = k\}, \quad S_H(k) = \#\{\pi : T_H > k\}.$$

The at-risk population just before step  $k$  is

$$A_H(0) = 16!, \quad A_H(k) = S_H(k-1) \quad (1 \leq k \leq 16).$$

This distinction between  $T_H \geq k$  and  $T_H > k$  fixes the boundary convention. Whenever  $A_H(k) > 0$ , the discrete hazard and transition survival are

$$h_H(k) = \Pr(T_H = k \mid T_H \geq k) = \frac{N_H(k)}{A_H(k)}, \quad 1 - h_H(k) = \frac{S_H(k)}{A_H(k)}.$$

If the at-risk population is empty, neither conditional probability is defined and the receipt records `null`.

## 2 Conditional residual-life theorem

After survival through step  $k$ , define

$$R_{H,k} = T_H - k \mid T_H > k.$$

Thus  $R_{H,k} \geq 1$  whenever it is defined.

**Theorem 1.** *For each of the twenty targets, each  $0 \leq k \leq 16$  with  $S_H(k) > 0$ , and  $0 \leq r \leq 16 - k$ , the C94 receipt contains the exact identities*

$$\begin{aligned} \Pr(R_{H,k} > r) &= \frac{S_H(k+r)}{S_H(k)}, \\ \Pr(R_{H,k} = r) &= \frac{N_H(k+r)}{S_H(k)} \quad (1 \leq r \leq 16 - k), \\ m_H(k) &:= \mathbb{E}[R_{H,k}] = \sum_{r=0}^{15-k} \frac{S_H(k+r)}{S_H(k)}, \\ q_H(k) &:= \mathbb{E}[R_{H,k}^2] = \sum_{r=0}^{15-k} (2r+1) \frac{S_H(k+r)}{S_H(k)}, \\ v_H(k) &:= \text{Var}(R_{H,k}) = q_H(k) - m_H(k)^2. \end{aligned}$$

If  $S_H(k) = 0$ , no conditional law or moment is asserted.

*Proof.* On  $\{T_H > k\}$ , the event  $R_{H,k} > r$  is exactly  $T_H > k + r$ , and  $R_{H,k} = r$  is exactly  $T_H = k + r$ . Division by the nonzero conditioning count proves the first two formulas. For an integer-valued nonnegative variable  $R$ , the pointwise identities

$$R = \sum_{r \geq 0} \mathbf{1}_{R > r}, \quad R^2 = \sum_{r \geq 0} (2r + 1) \mathbf{1}_{R > r}$$

give the two finite tail sums. The variance formula follows directly.  $\square$

**Proposition 1.** *The hazards determine the survival law recursively:*

$$S_H(k) = 16! \prod_{j=0}^k (1 - h_H(j))$$

through every defined transition. Conversely,  $N_H(k) = A_H(k) - S_H(k)$ , so the C94 hazard atlas and the frozen C88 distribution contain equivalent finite information once the boundary convention is fixed.

*Proof.* The identity  $S_H(k) = A_H(k)(1 - h_H(k))$  starts with  $A_H(0) = 16!$  and uses  $A_H(k) = S_H(k - 1)$  thereafter. The converse is the same identity rearranged.  $\square$

### 3 Exact inventory

The receipt has  $20 \cdot 17 = 340$  hazard steps and two complete  $20 \cdot 17 \cdot 17 = 5780$  residual grids, one for survival and one for mass. There are 261 nonempty conditioning rows and 2709 defined cells in each residual grid. Structurally impossible cells  $r > 16 - k$  and cells following an empty conditioning event are `null`, rather than assigned a numerical probability.

Representative exact values illustrate the range of laws:

| target | order | $m_H(0)$   | $v_H(0)$             |
|--------|-------|------------|----------------------|
| 1      | 2     | 17/2       | 85/4                 |
| 2      | 3     | 3961/1320  | 3586439/1742400      |
| 13     | 9     | 34/9       | 21488/4455           |
| 19     | 54    | 36499/3960 | 1652162153/109771200 |

The trivial target 0 has  $T_H = 0$  identically, so  $h_H(0) = 1$  and the event  $T_H > 0$  is empty; accordingly its residual row at  $k = 0$  is undefined.

### 4 Independent audit and scope

The producer reads only the canonical C88 evidence and manifest with SHA-256 digests

4511d434f477784782f2af5106afff4c2cf3b48cd7eb7a62ed05b8f2f42afb1b,  
aab137987b45be54d401b5a021212412de25097b149a73ee65c8e0daaced56c5.

The independent checker starts from each C88 hit bitset, enumerates pivotal edges to reconstruct  $N_H(k)$ , and then rebuilds the entire expected receipt. SymPy verifies all 340 hazards, 5418 defined survival/mass cells, and 261 defined mean–variance rows. Clean replay preserves evidence digest

e185462629459a7d6602e3d1e3f49977a82d3fdee86007c3f906b224f028d1b3.

All thirteen semantic mutations are rejected.

No arithmetic/local-data, Euler-factor, root-number, automorphy, full Burnside-ring/table-of-marks, or Hilbert–Polya operator claim is made.